ज्ञानेन प्रकाशते जगत्
INDUS UNIVERSITY
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NAAC ACCREDITED

ADAPTIVE INVENTORY MANAGEMENT SYSTEM

Branch : B. Tech CSE
Group Id : IITE/CSE2026/UDP088

By,
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2. Literature Review / Comparisons with Existing System
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4. Problem Statement
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1 Objective

- The objective of this project is to design and develop a user-friendly web-based system that enables users to create and manage customized online databases without requiring technical or database-specific knowledge.

- The system aims to provide a flexible and organized platform for storing, maintaining, and retrieving personal as well as organizational data such as documents, expenses, inventories, and records.

- By allowing users to define their own data fields and automatically generating structured forms, the project simplifies data entry and management while reducing dependency on traditional spreadsheets or manual record-keeping methods. Additionally, the system focuses on ensuring secure access, controlled data usage, and efficient data organization, making it suitable for both individual and professional environments.

2 Literature Review/Comparisons with existing system

Feature	Existing System	Proposed System
Method	Manual or rule-based inventory tracking using spreadsheets or basic ERP systems. Inventory decisions depend on fixed thresholds and human judgment.	Dynamic inventory system allows the user to drag and drop the provided fields according to their needs which allows flexibility and also take the help of AI based chatbot for smart queries.
User personalization	Same logic applied to all users regardless of business type.	Personalized inventory insights per user and per inventory using user-specific data.
Speed	Slow and time-consuming due to manual updates and delayed reporting..	Fast processing with real-time inventory insights and instant recommendations.
Adaptability	Limited flexibility; difficult to handle demand fluctuations or new product categories.	Highly adaptive, as user itself decides the schema of its inventory and flood it with require data which ignores the static forms.
Scalability	Not easily scalable	Easily scalable with more data and fields.
Cost	High operational cost due to labor, inefficiencies, and overstock/understock issues.	Cost-efficient in the long run by reducing waste, optimizing stock levels, and minimizing manual intervention.

3 Drawback of existing system

Existing data management systems such as spreadsheets and traditional database applications provide limited flexibility, require technical knowledge, and become difficult to manage as data grows. Fixed database structures make it hard for non-technical users to modify or customize data fields, while many no-code platforms are either complex, expensive, or lack integrated document handling and intelligent features.

The proposed system overcomes these limitations by allowing users to create **fully customized databases with dynamic fields** and **auto-generated forms** through a simple interface. In addition, the system enhances usability through AI chatbot features such as **Natural language search**,**automatic data insights**, making **data organization**, **retrieval**, and **management** more efficient compared to existing systems.

4 Problem Statement

Traditional inventory management systems are rigid, schema-dependent, and require technical expertise to extract insights, making them unsuitable for diverse and evolving business needs. They fail to adapt to dynamic inventory structures, struggle with the cold-start problem in demand prediction, and lack intelligent interfaces for real-time decision support. There is a need for an adaptive, AI-driven inventory management system that supports dynamic schema creation, natural language data querying, intelligent stock prediction, actionable analytics, and location-based supplier discovery to enable efficient, scalable, and data-driven inventory management.

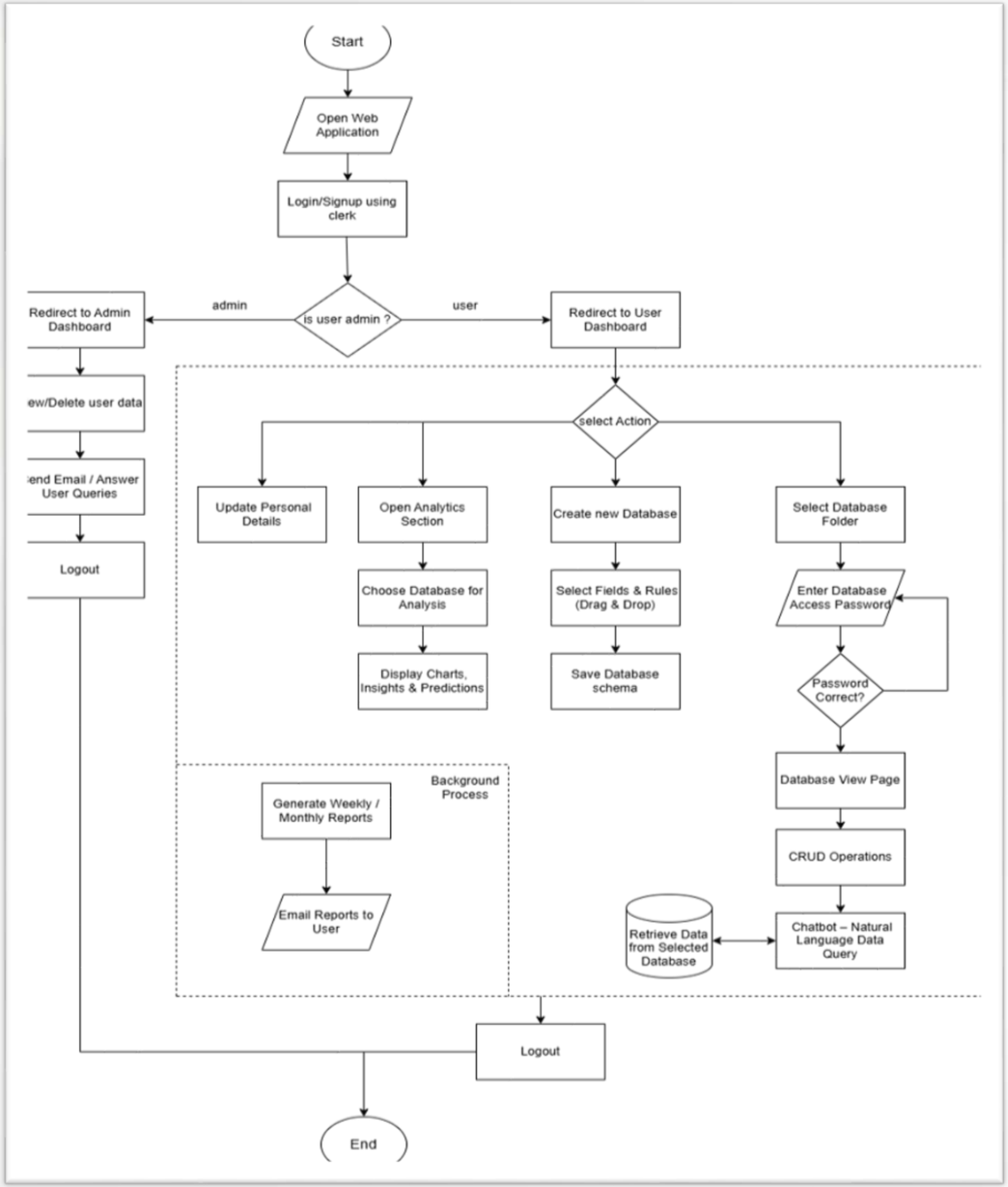
5 Research Gap

- Existing inventory management systems primarily rely on static database schemas, rule-based stock monitoring, and manual query mechanisms, which limit their adaptability across diverse business domains and user requirements.

- Most conventional systems also assume the availability of sufficient historical data for forecasting, making them ineffective for new or small-scale users facing the cold-start problem.

- Additionally, current solutions often lack intelligent interfaces for natural language interaction and do not provide seamless integration of predictive analytics, real-time insights, and location-based supplier discovery within a unified platform.

- There is a significant research gap in developing a scalable, adaptive inventory management system that supports dynamic schema creation, intelligently transitions from rule-based to data-driven predictions, enables natural language data access, and offers actionable analytics along with nearby supplier recommendations, particularly for small and medium-scale enterprises.

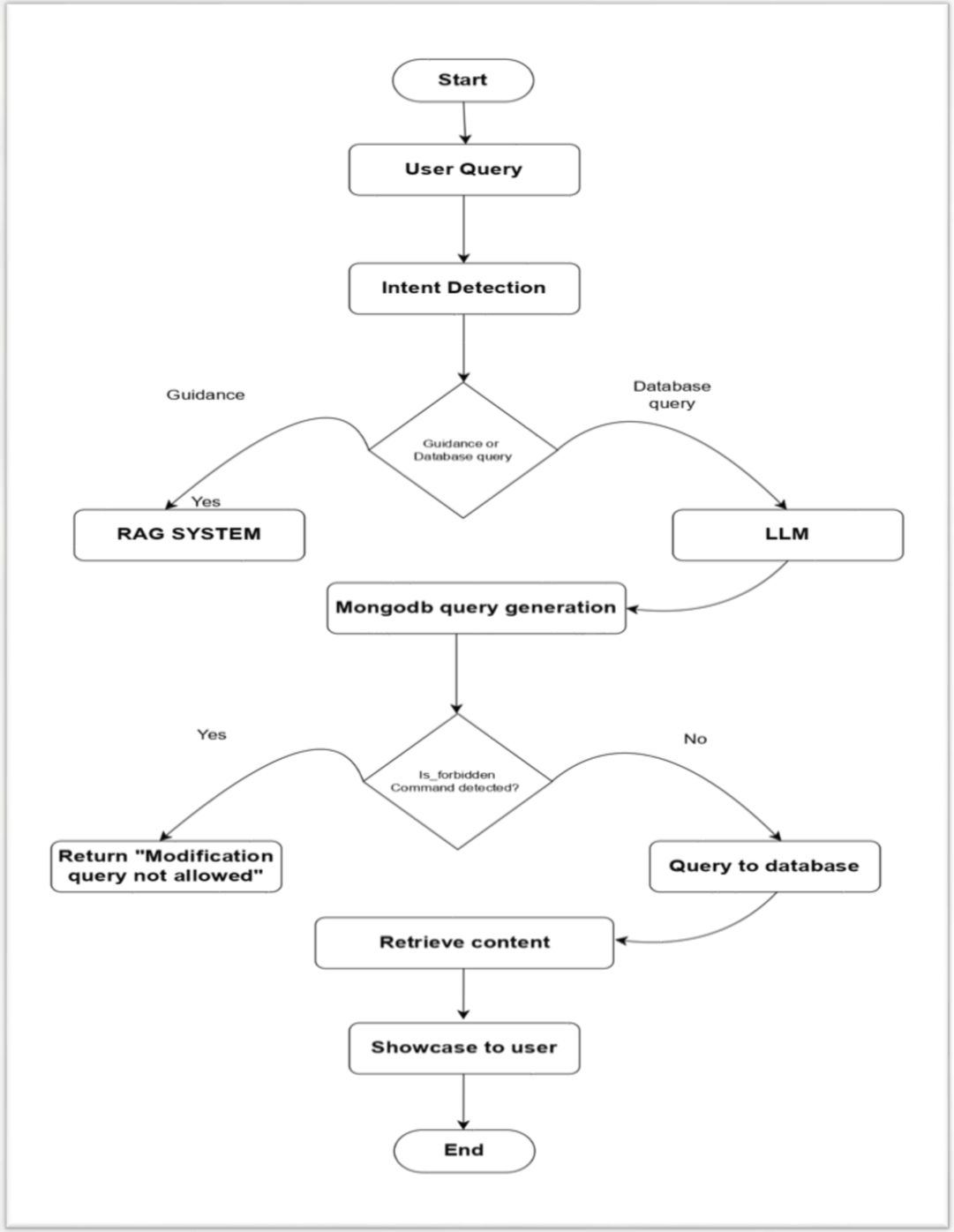


PROJECT WORKFLOW

1. Dataset Preparation

- 5000+ welding images
- Annotated in YOLO Format

- This is the **First Phase** of the project which consist of data collection from various sources including the Kaggle,google,etc.
- More than **5000+ images** are collected from the various sources.
- Further, the collected image is annotated using **the RoboFlow software** tool.
- The annotation include the labelling of the defects and creating the bounding box.



CHATBOT WORKFLOW

2. Model Training (YOLOv8)

- Splitiing into train,test and validation
- Training using Ultralytiics in Colab

- This is the **Second Phase** of the project which consist of **Yolov8** model training.

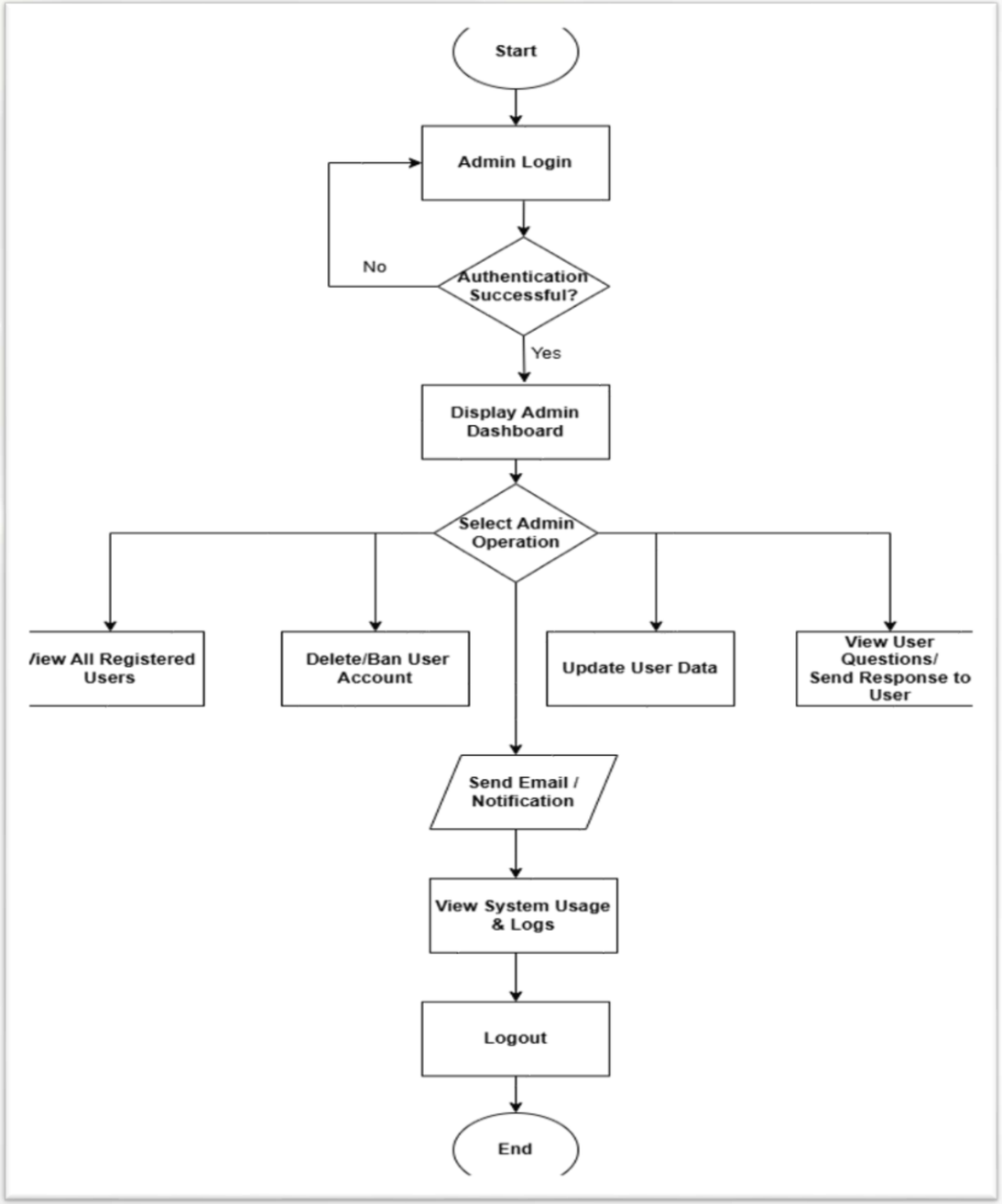
- In this process we first divide the image dataset into three parts i.e train,test and validation.

Train: Around **4000+** images has been used for training(tentative).

Validation: Around **1000+** images has been used for validation(tentative).

Testing: Around **10+** images has been used for testing(tentative).

ADMIN WORKFLOW



3. Defect Detection & Classification

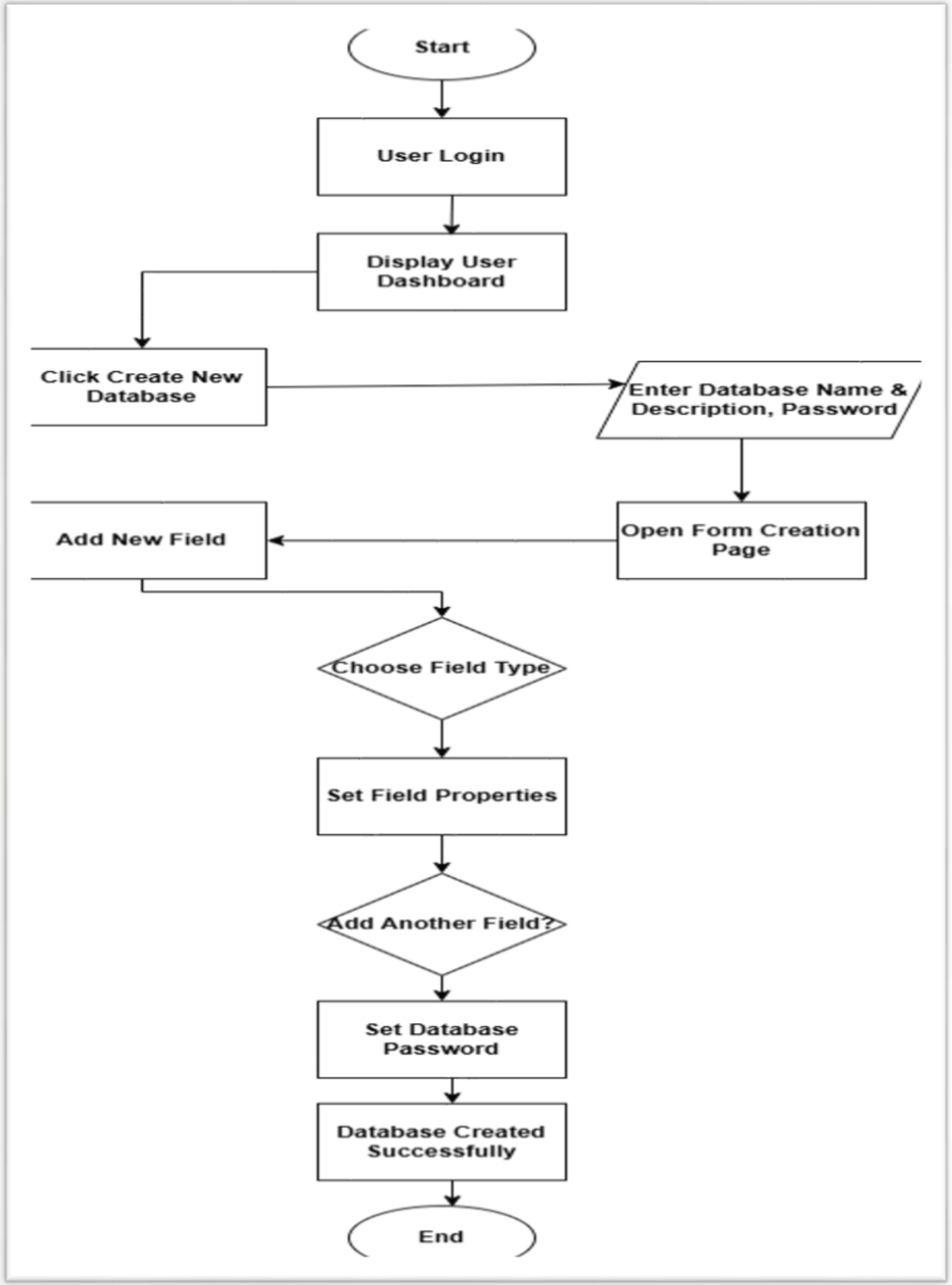
- The trained model detects the defects present in the image
- It classfies defects bounding box

- This is the **Third Phase** of the project. In this Phase the model is validated by providing the testing images so in order to check the accuracy of the model.

- The model consist of different evaluation metrics in order to determine the performance of models such as **precision,recall,mAP score.**

- Further in this phase the model takes the images and detects the defect present in it based in trained model and classify the defect labels.

DATABASE CREATION WORKFLOW



4. Creation of metadata.csv

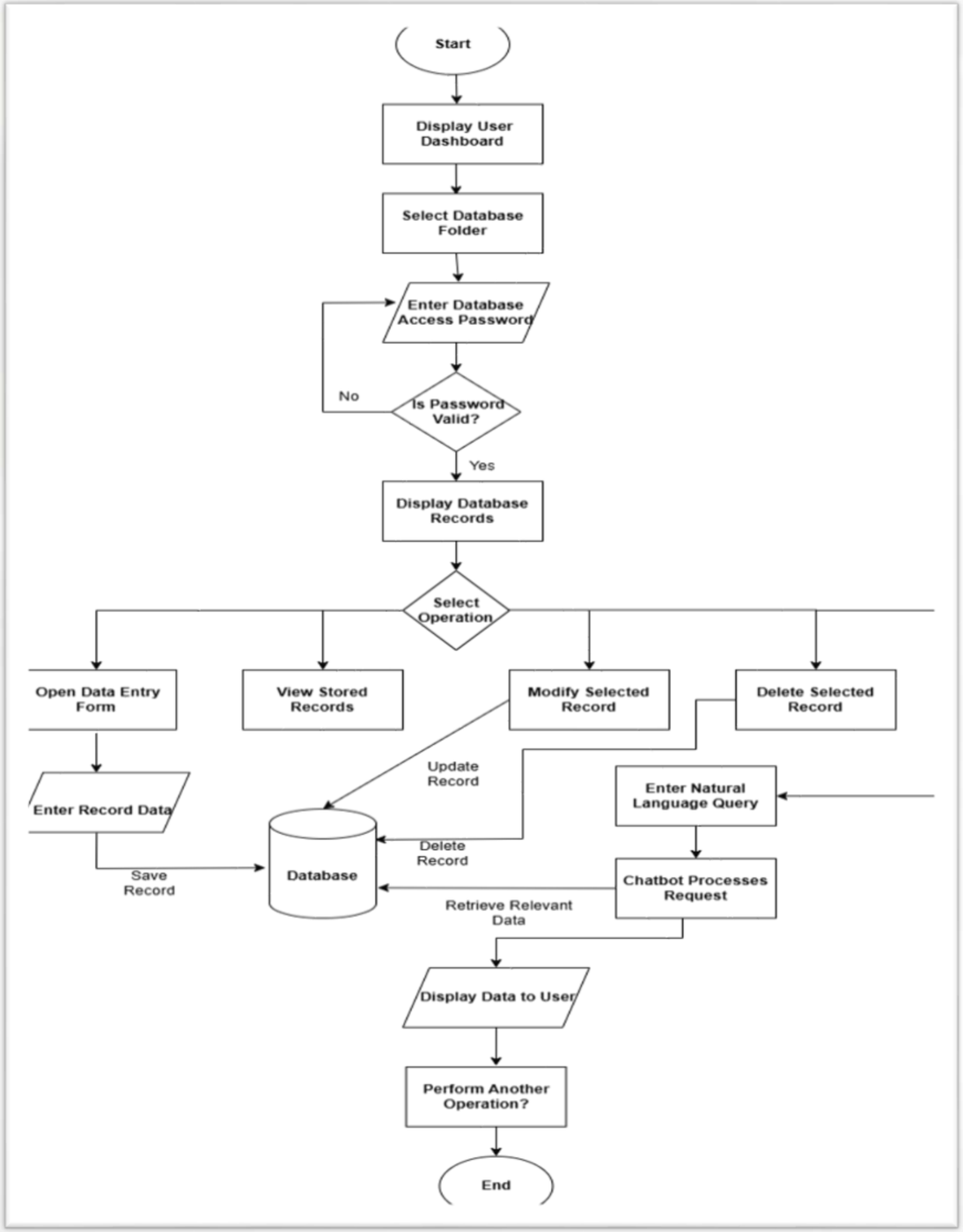
- Writing the logic to create the csv file which consist of metadata.

- This is the **Fourth Phase** of the project. In this Phase,we are going to create the **metadata** (Data About the data).

- The metadata consist of the **image_id,defect_type,no_of_defects,severity ,manufacture_date,remaining_life_span,rem arks.**

- This metadata.csv file helps in making the prediction model.

DATABASE ACCESS WORKFLOW



5. Prediction of lifespan

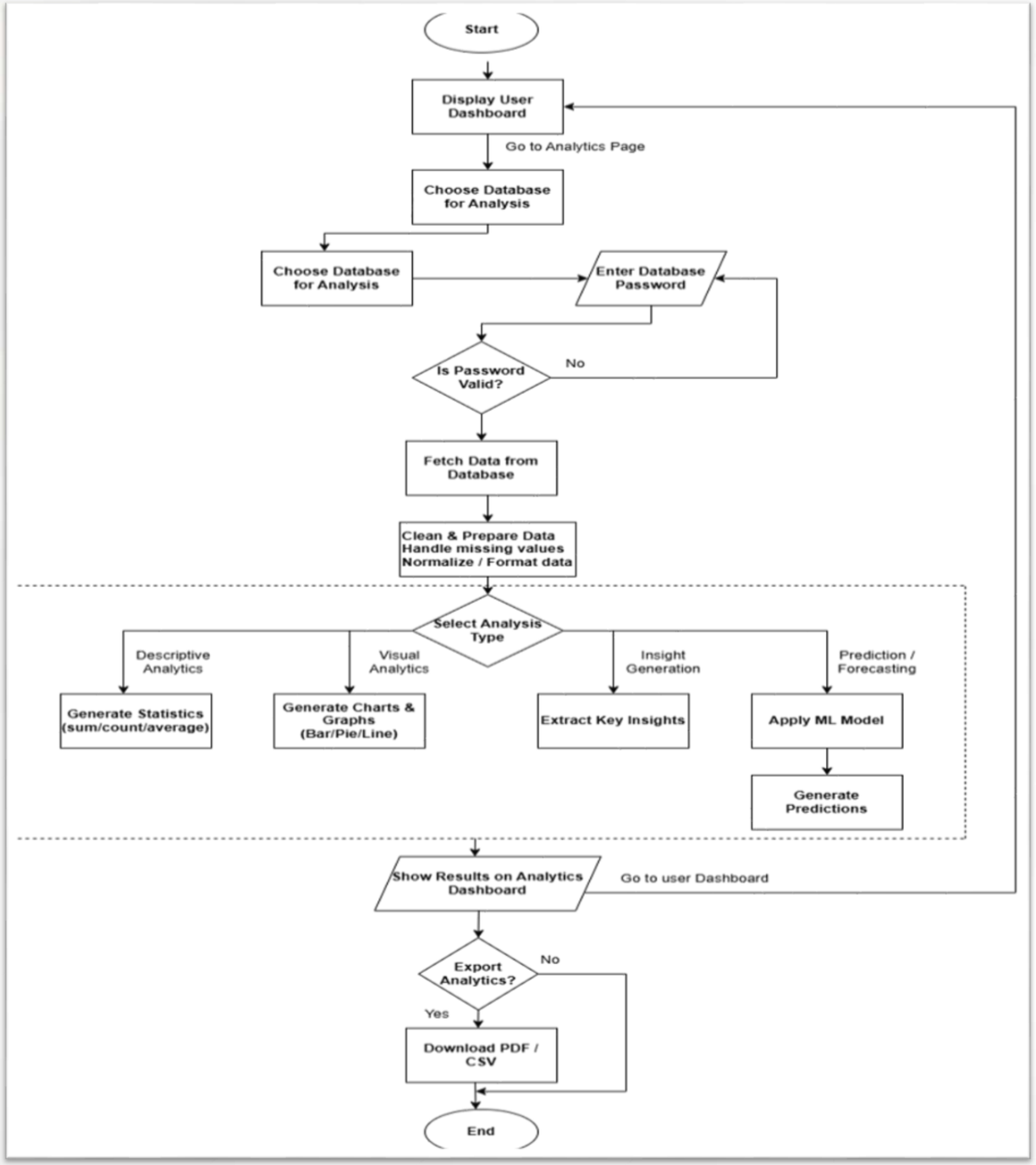
- Trained the regression model which predicts the remaining life of object.
- It uses the severity,no_of_defects and type of defects as main feature.

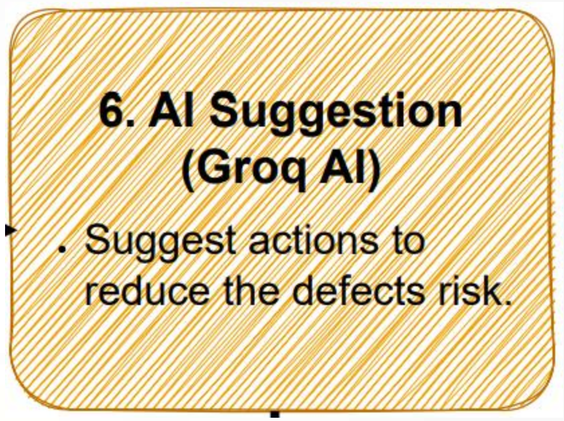
- This is the **Fifth Phase** of the project. In this Phase, we are going to create the **Prediction Model**, which is trained on various features provided inside the metadata.csv file.

- The main features which helps the model to predict the **remaining_life_span** is **defect_type,no_of_defects and severity.**

- This model is crucial in order to determine the life span of the object which ultimately help the manufacturer to take actions.

ANALYTICS WORKFLOW





- This is the **additional feature** which provides the customer suggestions based on the defect detected.
- This phase includes the integration of **Groq AI** which provides the measures in order to reduce the cause of welding defects.
- This helps the industrialist to take precautions during the welding process so that minimum defects should arise.

6 Social Benefits

1. Support for Small and Local Businesses

Small retailers and shop owners can manage inventory efficiently without advanced technical knowledge, helping them compete with large enterprises.

2. Reduction in Manual Errors

Automated stock tracking using **insights** and **AI Chatbot** minimize human errors in inventory recording, leading to more reliable business operations.

3. Employment Productivity Enhancement

Employees spend less time on manual stock checking and more time on customer service and business growth, improving workplace efficiency.

4. Scalability Across Multiple Sectors

The adaptive schema design allows usage across pharmacy, retail, hardware, and other sectors, promoting digital transformation in diverse industries.

5. Faster Stock Replenishment

The **“nearby store locator”** enables users to quickly identify local suppliers, reducing procurement time during stock shortages.

6 Social Benefits

6. Eliminates the Need for Technical Query Knowledge

Users can retrieve inventory data without knowing SQL or database schemas, making the system accessible to non-technical users.

7. Supports Dynamic and Adaptive Schemas

Since users can define their own inventory fields, the chatbot dynamically maps queries to evolving schemas without requiring rigid predefined forms.

8. Clear Visibility into Stock Movement

Users gain real-time insights into inventory inflow, outflow, and current stock levels..

9. Revenue and Profit Optimization

Sales and pricing trends assist in identifying opportunities to improve margins and profitability.

7 Used Technology

FRONTEND :





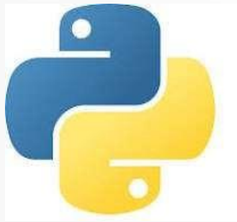








BACKEND :







DATABASE:



TIMELINES

Adaptive Inventory Management System

1st Jan - 28th March



8 List of Modules

Module 1: Authentication Module (Sign In / Sign Up)

Sub-Modules:

1.1. User Registration
1.2. User Login Authentication
1.3. Role-Based Access Control (Admin / User)
1.4. Password Encryption & Security
1.5. Session Management
1.6. Logout & Token Expiry Handling

Module 2: User Dashboard Module

Sub-Modules:

2.1. Inventory Overview Summary
2.2. Low Stock Alerts & Notifications
2.3. Quick Action Panel
2.4. Recent Activity Tracking
2.5. Personalized Insights Display
2.6. Profile Management

8 List of Modules

Modules 3: Admin Dashboard Module

Sub-Modules:

3.1. User Management (View, Activate, Deactivate Users)
3.2. Inventory Category Management
3.3. System Usage Monitoring
3.4. Data Access Control
3.5. Platform Analytics Overview
3.6. Role & Permission Management

Module 4: Inventory Management Module

Sub-Modules:

4.1. Dynamic Schema Builder (Drag & Drop Fields)
4.2. Automatic Form Generation
4.3. Product Creation
4.4. Product Update
4.5. Product Deletion
4.6. Bulk Product Upload (Excel / CSV)
4.7. Inventory Validation & Constraints
4.8. Stock Level Management

8 List of Modules

Module 5: Product Filtering & Search Module

Sub-Modules:

5.1. Category-Based Filtering
5.2. Attribute-Based Filtering (Price, Quantity, Date, Custom Fields)
5.3. Range-Based Filters
5.4. Keyword-Based Search
5.5. Dynamic Filter Adaptation (Based on User Schema)

Module 6: AI-Based Chatbot Module

Sub-Modules:

6.1. Natural Language Query Processing
6.2. Intent Detection & Classification
6.3. Dynamic Schema Mapping
6.4. Query Translation (NL → Database Query)
6.5. Secure Query Execution
6.6. Context-Aware Response Generation
6.7. Recommendation & Alert Generation

8 List of Modules

Module 7: Nearby Store Locator Module

Sub-Modules:

7.1. Inventory Category-Based Supplier Discovery
7.2. Location-Based Search
7.3. Distance Calculation
7.4. Map Visualization
7.5. Supplier Filtering & Ranking
7.6. External API Integration

8. Analytics & Insights Module (Final Analytics)

Sub-Modules:

8.1. Sales Trend Analysis
8.2. Demand Pattern Visualization
8.3. Stock Utilization Metrics
8.4. Product Performance Analysis
8.5. Forecasting Insights
8.6. Interactive Dashboards & Charts
8.7. Exportable Reports

9 Work Flow Diagram of Proposed System

10 Work Flow Diagram of Individual Modules For Proposed System

11 Next Phase Discussion Topics

Live Demonstration of the Following Modules:

Module 1: Authentication Module (Sign In / Sign Up)

Module 2: User Dashboard Module

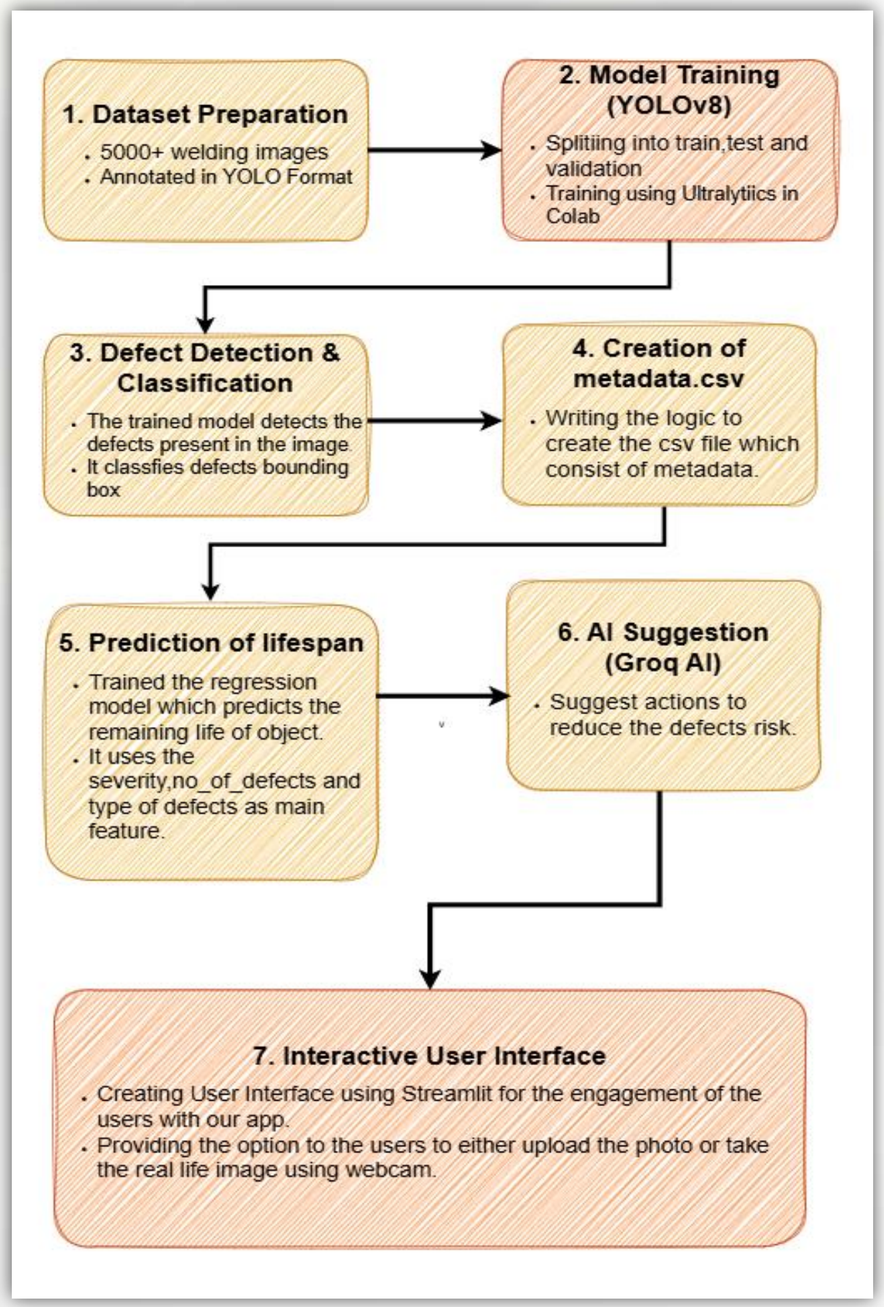
Modules 3: Admin Dashboard Module

Module 4: Inventory Management Module

Module 5: Product Filtering & Search Module

THANK YOU

PROJECT WORKFLOW



7. Interactive User Interface

- Creating User Interface using Streamlit for the engagement of the users with our app.
- Providing the option to the users to either upload the photo or take the real life image using webcam.

- This is the final phase of the project which includes the creation of the **Interactive User Interface** which help the consumers to interact with the UI.
- The UI is created using the **Streamlit** Library which helps in effectively creating the UI components such as buttons,text field,upload options,etc.
- The UI Includes the options for the users to either **Upload** the Photo of their Welding Component or directly **Capture** the picture.